

Probability

Experiment - an action you ^{do} or take to see the result of that action (outcomes of the actions)

There are 2 types of experiments:

* Random

* Non-random

Non-random - you are certain of the outcome before you do the experiment

Random - you know all possibilities of outcomes before experimentation but you only know the actual outcome once the experiment is performed.

you don't know which will be the outcome before experiment.

Sample space (Ω) - the collection of all possible outcomes of an experiment

they are written in set form.
{ }

i.e. When tossing a coin - experiment
 $\Omega = \{\text{head}, \text{tail}\}$

Event - subset of a sample space.

↓
part of

given by 2^n

where n is the number
of elements in the set /
outcomes

power of
a set

i.e. $\{H\}$
 $\{T\}$
 $\{H, T\}$
 $\{\emptyset\}$

We normally denote events in capital letters
There are: simple events i.e. $\{H\}$
compound events

The probability of an event A is usually
described as

$$P(A) = \frac{\# \text{ of outcomes in } A}{\# \text{ of outcomes in sample space}}$$

Properties of a probability

$$0 \leq P(A) \leq 1$$

$$P(\emptyset) = 0$$

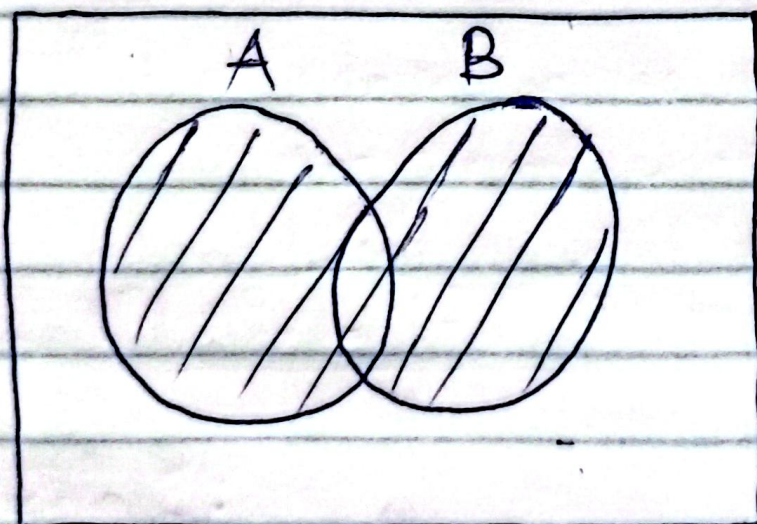
$$P(\Omega) = 1$$

Union of 2 events

$A \cup B$ or A or B

This means all elements in A or B or both

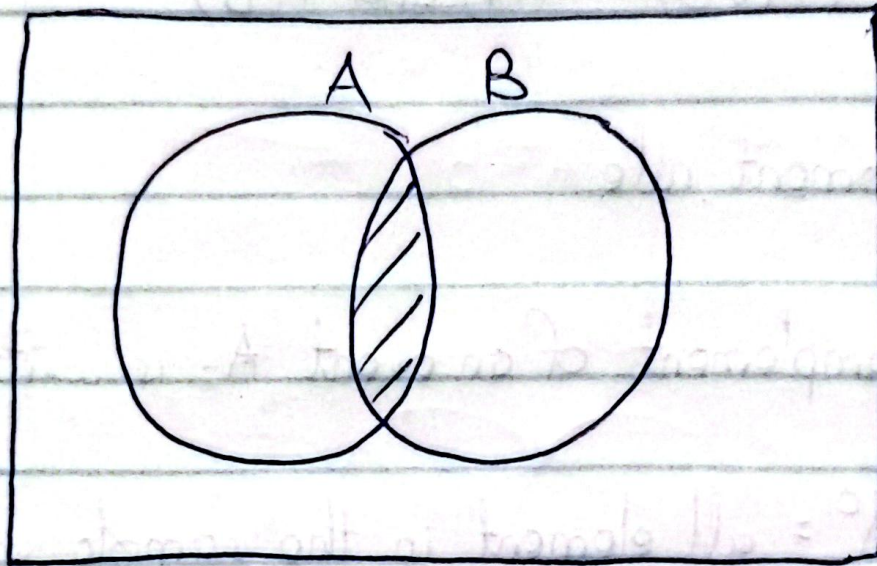
Using a venn diagram:



Intersection of 2 events

$A \cap B$ (A and B) - all elements that are in both A and B

Using a venn diagram:



Mutually exclusive events

If $A \cap B = \emptyset$, then A and B are mutually exclusive.

i.e. when you toss a coin. When the outcome is H the other possible outcome cannot happen. In this case we are talking about tossing a coin once OR

when you roll a die one outcome excludes all the other outcomes from happening

Rules for calculating probability

1. Addition rule

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are mutually exclusive

$$P(A \cup B) = P(A) + P(B)$$

2. Complement rule

The complement of an event A is written as A^c

A^c = all element in the sample space that are not in A

$$P(A^c) = 1 - P(A)$$

Example 1:

$$\Omega = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{1, 2, 4, 5\}$$

$$B = \{3, 5, 6\}$$

$$C = \{4\}$$

$$a). \text{ Prob } A \cup B = \Omega = \{1, 2, 3, 4, 5, 6\}$$

$$b). A \cap B = \{5\}$$

$$c). B \cap C = \emptyset$$

$$d). P(A \cup B) = 1$$

$$e). P(C^c) = 1 - P(C) = 1 - 1/6 = 5/6$$

Example 2: $P(B) = 1/8$ changed to $(1/2)$

Given $P(A) = 1/3$, $P(A \cup B) = 2/3$. Calculate:

$$a). P(A \cap B)$$

$$b). P(A^c)$$

$$c). P((A \cap B)^c)$$

$$P(A \cap B) = 1/3 + 1/8 = \frac{8+3}{24} = 11/24 - 2/3 = \frac{11-16}{24} = -5/24$$

incorrect
probability shouldn't
be $-5/24$

$$P(A \cap B) = 1/3 + 1/2 = \frac{2+3}{6} = 5/6 - 2/3 = \frac{5-4}{6} = 1/6$$

$$= 1/6$$

$$P(A^c) = 1 - 1/3 = 2/3$$

$$P((A \cap B)^c) = 1 - 1/6 = 5/6$$

De Morgan's law

$$(A \cup B)^c = A^c \cap B^c$$

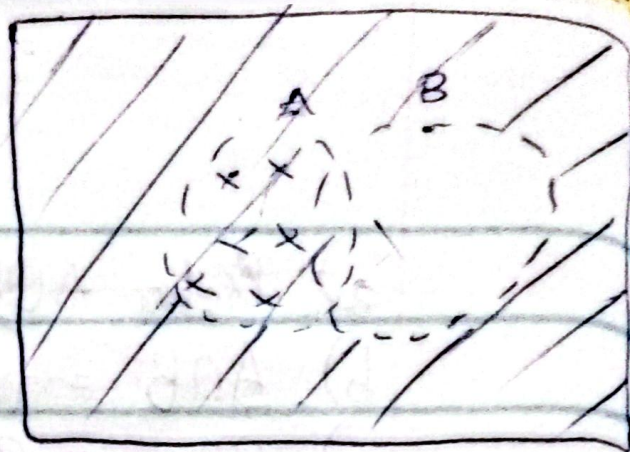
$$(A \cap B)^c = A^c \cup B^c$$

$$(A^c)^c = A$$

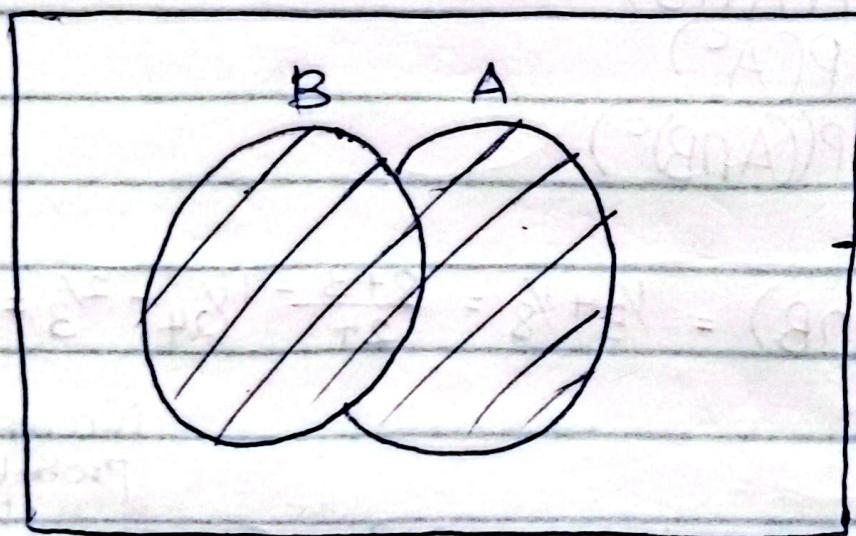
$$A = (A \cap B)^c \cup (A \cap B)$$

A or B but not both

$$(A \cap B^c) \cup (A^c \cap B)$$



used to
explain



universal
set

Conditional probability ^{ies} → introducing a condition
↓
changes the sample space.

All probabilities are usually conditional probabilities

$$P(A) = P(A|\Omega)$$

Let's say you have $P(A|B) = \frac{\text{\# of elements in } A \cap B}{\text{\# of elements in } B}$

$$\frac{P(A \cap B)}{P(B)}$$

probability of B given A

$$P(B|A) = \frac{P(B \cap A)}{P(A)}$$

$$P(B \cap A) = P(A \cap B)$$

$$P(B \cap A) = P(B|A) \cdot P(A)$$

$$P(A \cap B) = P(A|B) \cdot P(B)$$

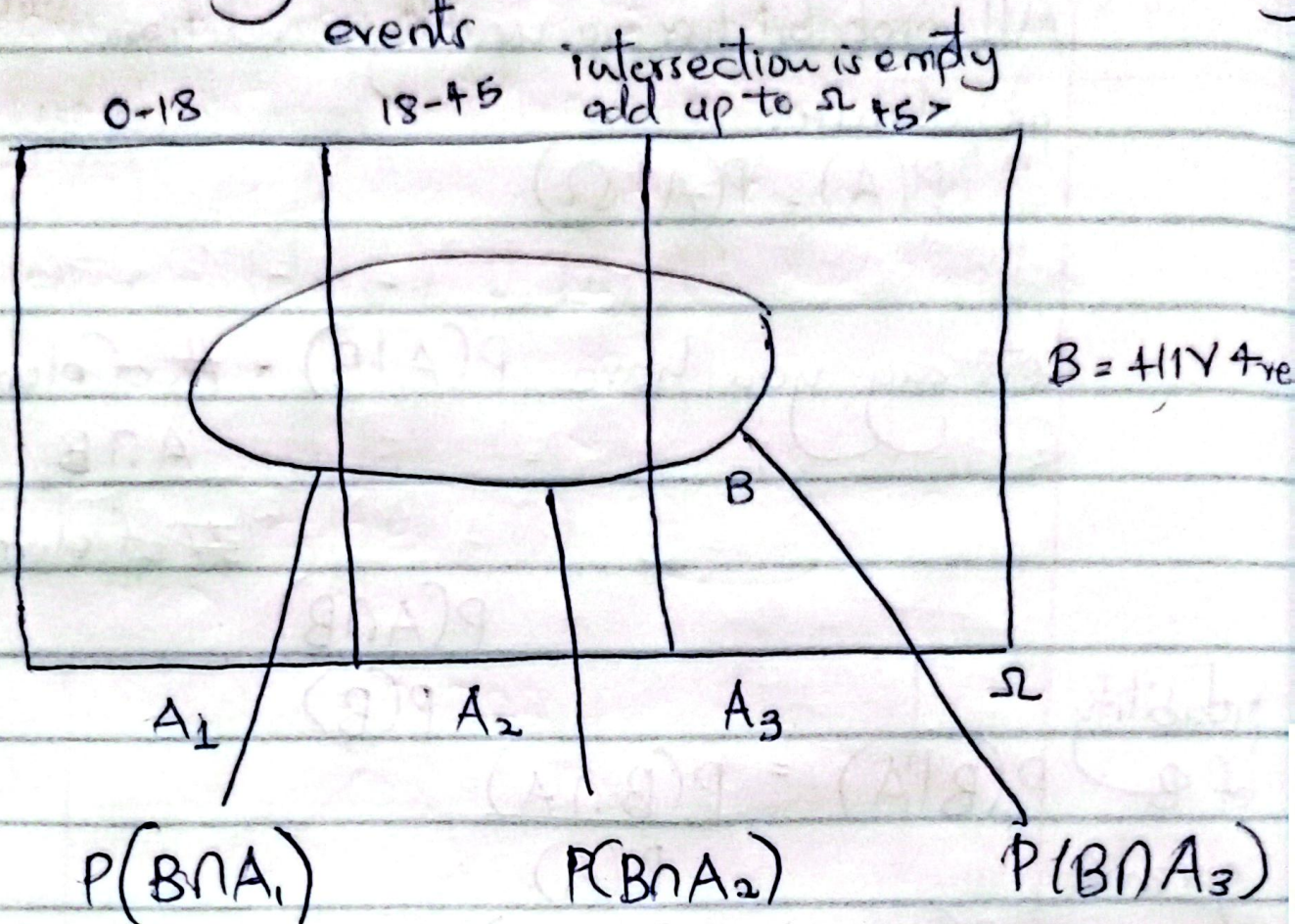
$$P(B|A) \cdot P(A) = P(A|B) \cdot P(B)$$

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)}$$

Bayes' Theorem/Theorem - helps you to invert the conditional probability

people who can swim are females
 females given they can swim start with picking females
 swimmers

partitioning - divide sample space into non-overlapping events



$$P(B) = P(B \cap A_1) + P(B \cap A_2) + P(B \cap A_3)$$

$$= P(B|A_1) \cdot P(A_1) + P(B|A_2) \cdot P(A_2) + P(B|A_3) \cdot P(A_3) \dots$$

Partition theorem (Law of total probability)

Example 1:

Given $P(A) = \frac{1}{3}$, $P(B) = \frac{1}{2}$ and $P(A|B) = \frac{1}{8}$

Calculate probability of $(B|A)$ B given A

$$P(B|A) = \frac{P(A|B) \cdot P(B)}{P(A)} \quad \frac{\frac{1}{8} \cdot \frac{1}{2}}{\frac{1}{3}} = \frac{\frac{1}{16}}{\frac{1}{3}} = \frac{3}{16}$$

Probability of getting a girl or a boy is equally likely. In a family of 2 children what is the probability of getting a girl given ^{your already} the first child is a boy

$$P(B) = \frac{1}{2}$$

$$P(G) = \frac{1}{2}$$

$$\Omega = \{BB, BG, GB, GG\}$$

$$\{BB, BG\}$$

$$\{BB, BG, GB\}$$

$$P(G) = \frac{1}{2}$$

$$P(B) = \frac{1}{2}$$

$$P(G|B) = \frac{P(G \cap B)}{P(B)}$$

$$\frac{\frac{1}{2}}{\frac{3}{4}} = \frac{2}{3}$$

$$\frac{\frac{1}{2} \times \frac{4}{3}}{1} = \frac{2}{3}$$

In a population, the partitioning of a person having a disease is 0.05. The probability of a person testing positive and he has a disease is 0.99. The probability of testing positive given has no disease is ~~0.02~~ ^{0.002}. What is the probability the person ^{actually} has disease

$$P(D) = 0.05$$

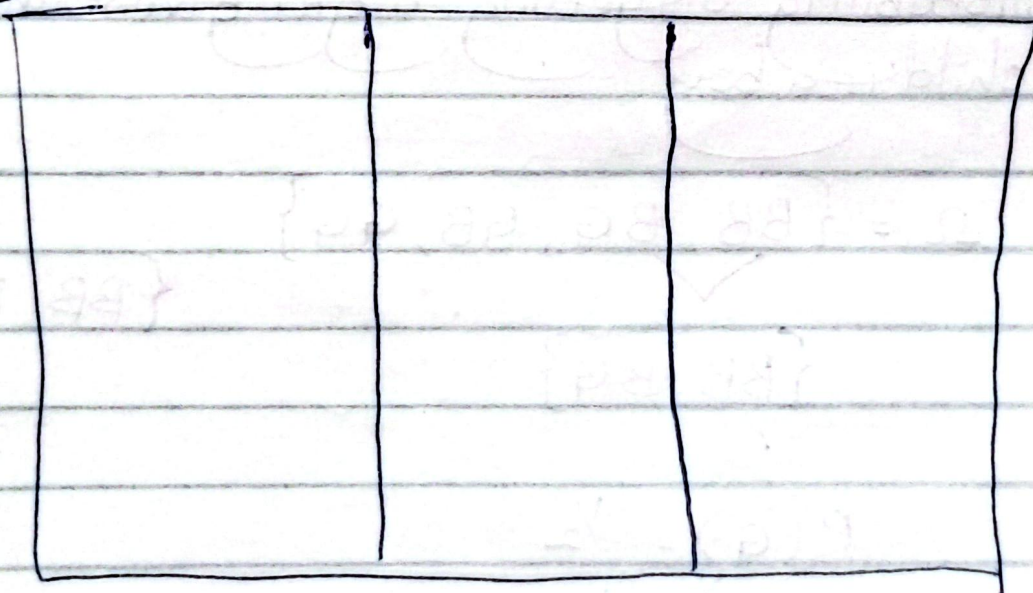
$$P(P^+|D) = 0.99$$

$$P(P^+|D^c) = 0.002$$

sensitivity of a test

general probability

diagnostic condition



Positive is when you $T^+|D$

starts
there who are

$$P(D) \cdot P(T^+|D) = \frac{\text{specificity}}{P(D)}$$

$$\frac{P(T^+|D) P(D)}{P(T^+)} = P(D|T^+)$$

$$\frac{0.99}{0.992} = 0.99798$$

$$P(T^+) = P(T^+/D) P(D) + P(T^+/D^c) P(D^c)$$

$$= 0.992$$

Independent events.

Are events that when one occurs the probability of other one occurring is not affected

$$P(A|B) = P(A)$$

$$P(A \cap B) = P(A) \cdot P(B)$$

only true for independent events.

Multiplication rule

$$P(A \cap B) = P(A|B) \cdot P(B)$$

Random Variables.